

```
From Stdlib Require Import List.  
Import ListNotations.
```

Recall the definition of the type of lists with entries in a type A.

```
Inductive list (A : Type) : Type :=  
| nil : list A  
| cons : A → list A → list A.
```

Arguments nil {A}.

Arguments cons {A} a l.

```
Infix "::-" := cons (at level 60, right associativity) : list_scope.
```

Recall also the definition of the function negb.

```
Definition negb (b : bool) : bool :=  
  match b with  
  | true => false  
  | false => true  
  end.
```

Exercise 1

Write a function `length : list A → nat` that returns the length of a list.

Exercise 2

Prove the following result:

```
Theorem negb_inv (b : bool) : negb (negb b) = b.
```